



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to ARUN FUEL

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

#### Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

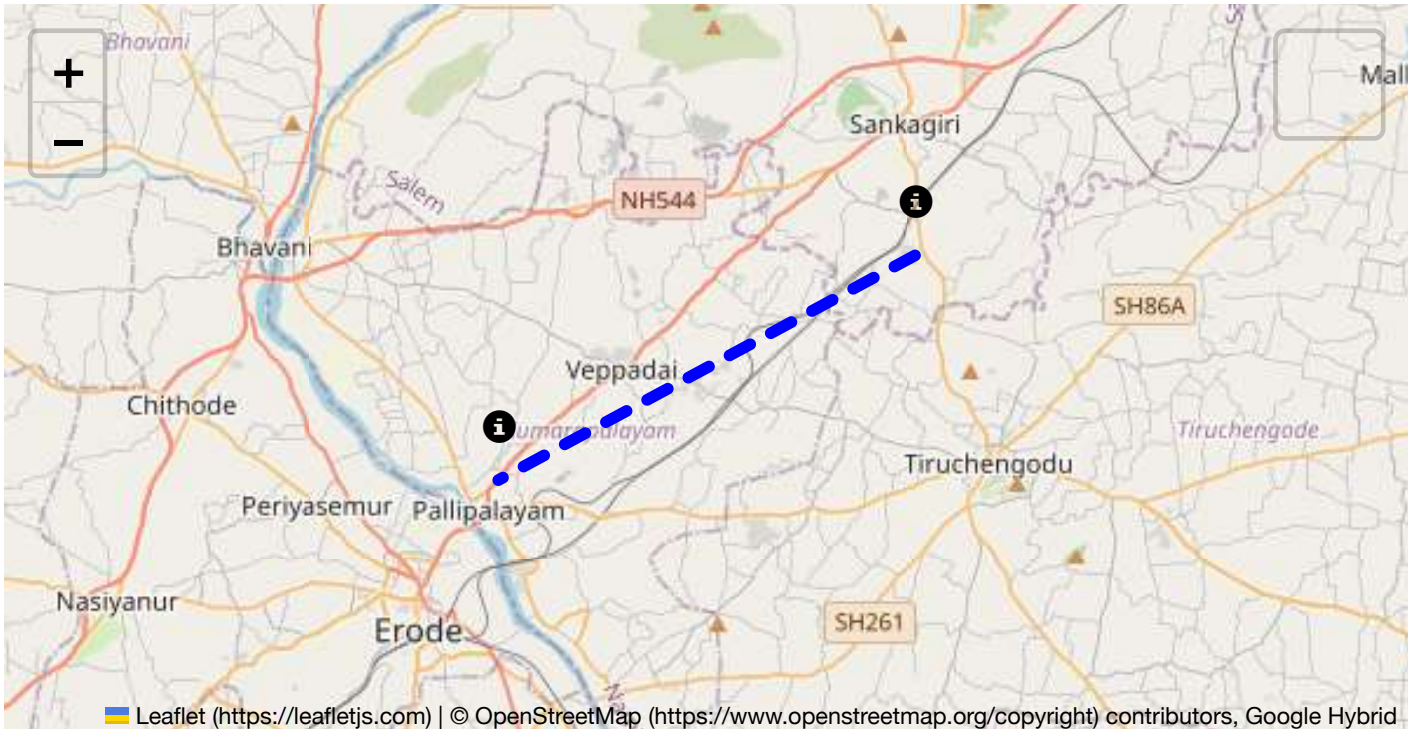
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



**Route Summary:**  
**Total Distance: 20.51 km**  
**Estimated Duration: 0.5 hours**  
**Adjusted Duration (Heavy Vehicle): 0.6 hours**



End: (11.373918, 77.750817)



## 1. Overview of the Route Map

## 2. Typical Weather Conditions

- ### 3. Traffic Patterns

- ## 4. Assessment of Road Quality

- 3/8

## 5. Suggestions for Alternative Routes

- **Emergency Detours:** In case of roadblocks, consider taking state highways that parallel the main route but watch for local traffic advisories. Have GPS equipment updated for real-time traffic updates.

## 6. Local Regulations on Hazardous Material Transport

- **Regulations:** Transport of hazardous materials is subject to permits and specific routes designated by local authorities. Check with Tamil Nadu Pollution Control Board for current guidelines.

## 7. Historical Incidents

- **Recorded Incidents:** There have been instances of vehicular accidents on highways, primarily due to overloading or speeding. However, specific data on hazmat incidents may be limited without direct regional sources.

## 8. Environmental Considerations

- **Sensitive Areas:** There are no major ecological reserves directly along the route. However, be cautious near agricultural zones to prevent contamination risks.

## 9. Communication Coverage

- **Network:** Mobile coverage is generally reliable on major parts of the highway but may drop in rural sections.
- **Dead Zones:** Potential dead spots at low-lying areas and narrow valleys—ensure alternate communication means like a satellite phone.

## 10. Estimated Emergency Response Times

- **Urban Areas:** Response times are around 15-20 minutes due to proximity to service facilities.
- **Rural Sections:** Times might extend to 30-45 minutes. It is advised to have emergency contacts pre-programmed into communication devices.

## 11. Overall Risk Assessment

- **Summary:** The route is relatively short, but potential risks include traffic congestion, weather-related challenges during monsoons, and road quality issues on minor roads. Adequate preparation, including route verification and adherence to regulations, can mitigate most risks.

In conclusion, while the route presents typical challenges expected in semi-urban and rural conduits, adherence to safety protocols and preemptive planning can ensure safer transportation of hazardous materials. Regular updates on traffic and weather, coupled with clear communication channels, are crucial for navigating these risks effectively.

|   | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|---|------------|------------|--------------------|-------------|
| 0 | Roundabout | High       | 11.37086, 77.74828 | 15 KM/Hr    |
| 1 | Turn       | High       | 11.43811, 77.87348 | 15 KM/Hr    |
| 2 | Turn       | Medium     | 11.43956, 77.87340 | 30 KM/Hr    |
| 3 | Turn       | Medium     | 11.43971, 77.87348 | 30 KM/Hr    |
| 4 | Turn       | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 5 | Turn       | Medium     | 11.46307, 77.84941 | 30 KM/Hr    |
| 6 | Turn       | Medium     | 11.46318, 77.84913 | 30 KM/Hr    |
| 7 | Turn       | Medium     | 11.46312, 77.84886 | 30 KM/Hr    |
| 8 | Turn       | High       | 11.37391, 77.75093 | 15 KM/Hr    |

Emergency Locations

Crowded Spots

|   | type   | name                                      | coordinates            | speed_limit | risk_level |
|---|--------|-------------------------------------------|------------------------|-------------|------------|
| 0 | school | Ssri Valliappa Vidhayalayam mat hr school | 11.4004878, 77.7794381 | 30 km/h     | Medium     |

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

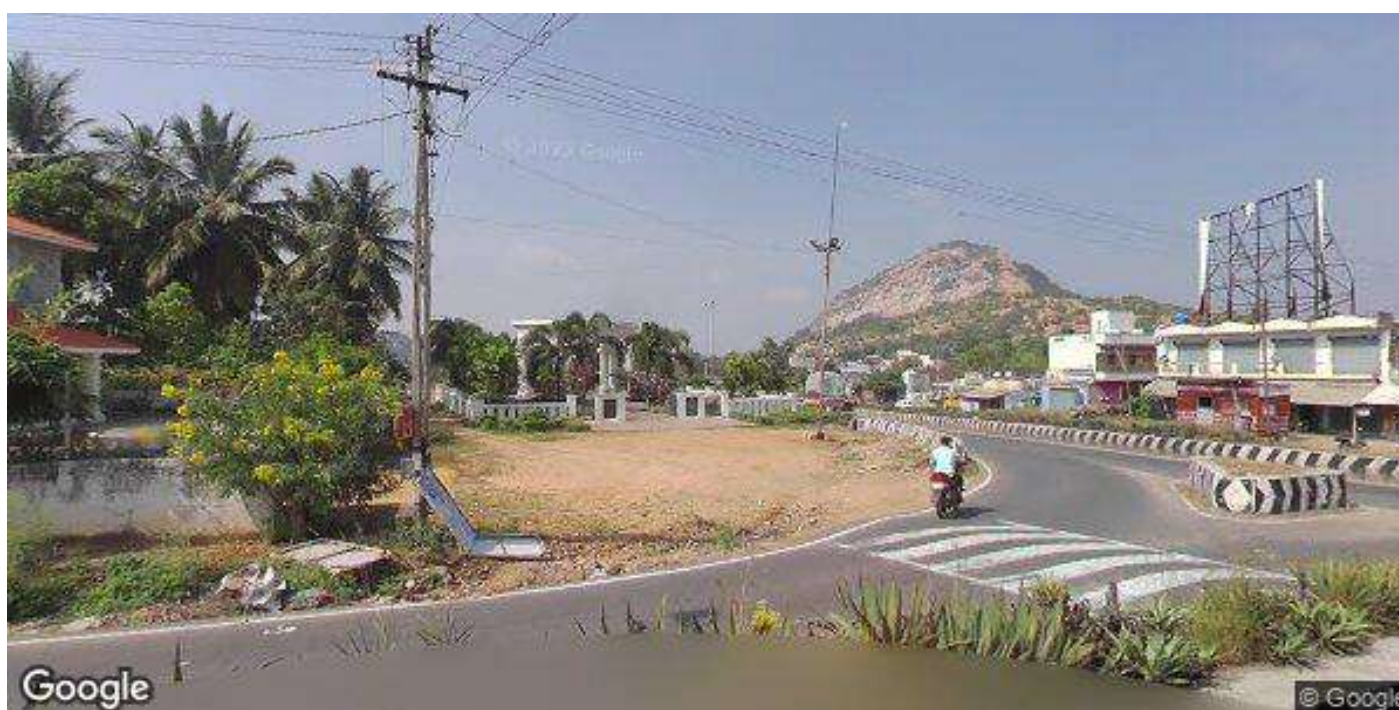
Speed Limit: 15 KM/Hr

Coordinates: 11.37086, 77.74828





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.44029, 77.87544



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46307, 77.84941



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46318, 77.84913





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46312, 77.84886



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.37391, 77.75093

